

SANJIVANAI

“A Personal AI Nurse for Every Family”

Remote health monitoring · on-time medicines · automatic emergency help — so families can care for elderly & chronically ill loved ones at home, even from far away.

Concept Document — Working Draft v1.0 (for team planning & submission prep)

Competition	Hack for Social Cause (HSC) 2027 — Youth Tech Challenge
Organizers	Ministry of Youth Affairs & Sports (MoYAS) · Knowledge Partner: IIT Bombay
Thematic areas targeted	Healthcare, Wellbeing & Service Delivery + Elderly Care & Healthy Ageing
Motto	“Think Global, Hack Local — Technology for Social Transformation”
Team	[Team name to be decided] — Team Lead, Developer, Content/Healthcare · State: West Bengal
Last date of submission	15 October 2026 (Registration open: 1 Sep – 15 Oct 2026)
National Showcase	VBYLD 2027, New Delhi — 10–12 January 2027
Status	Planning

1 Competition & Deadlines at a Glance

Item	Detail
Eligibility	Indian citizen; age 18–29 as on 17 August 2026 ; enrolled in an AISHE-registered college/university/polytechnic; team of up to 3 (same or different institutions; solo allowed).
Registration process	1) Every member registers individually on mybharat.gov.in → 2) Team Lead registers the team under "Hack for Social Cause" → 3) Team lead uploads all submissions.
Submission window	1 September – 15 October 2026
What to submit	Problem Statement · 6–7 slide Presentation Deck (≤10 MB, incl. AI-tools disclosure) · Working Prototype (public GitHub/GitLab) · Demo Video 3–5 min, ≥720p, ≤80 MB (preferred) · Team Self-Declaration + student ID (Annexure 1).
Stages	Stage 1: Registration & submission (by 15 Oct) → Stage 2: State hackathon (16 Oct–30 Nov) → Stage 3: IIT Bombay screening to 36 national finalists (1–15 Dec) → Stage 4: National Showcase, New Delhi (10–12 Jan 2027).
Evaluation (6 parameters)	Relevance · Technical Strength · Functionality · Creativity/Innovation · Social Cause Impact · Presentation & Team.
Prizes	1st –25,000 · 2nd –50,000 · 3rd –25,000 · 4th/5th –15,000 each · National finalists also get a development grant of –6,000 + mentorship by IIT Bombay.

2 The Problem: Why This Matters Today

Millions of Indian families quietly struggle with the same fear: an elderly parent or a chronically ill family member (high blood pressure, diabetes, heart disease, post-stroke care) needs round-the-clock attention... but the family has to work, children study, and life moves to the city. The result is a daily chain of small disasters:

- **Medicines are missed or doubled** — average medication adherence in India is below 50% for chronic conditions, leading to hospital re-admissions and complications (WHO reports poor adherence as a leading cause of avoidable deaths).
- **Danger signs are noticed too late** — a heart-rate or blood-pressure shift at 2 a.m. goes unnoticed until the morning, when it is a crisis instead of a warning.
- **Hospitals are treated as the default answer** — patients are admitted, routinely monitored, and discharged, while the family can rarely continue safe monitoring at home. Out-of-pocket health spending pushes families into debt.
- **Professional monitoring is unaffordable** — remote-patient-monitoring (RPM) devices and private care plans cost anywhere from ₹20,000 to lakhs per year — out of reach for most Indian households.
- **Emergency response has no bridge** — by the time a family member dials 108 or an ambulance, crucial minutes are already lost; there is no automatic, trustworthy path from a dangerous reading to the nearest emergency service.
- **Rural and non-English-speaking India is left out** — most monitoring apps are English-only, need stable internet, and ignore the elderly themselves, who are often the least comfortable with smartphones.

West Bengal context (our "Hack Local")

West Bengal has a rapidly ageing population (several districts among India's highest elderly ratios), a high burden of hypertension and diabetes, and a large rural population where a Primary Health Centre may be kilometres away. Working children often live in Kolkata or other metros while parents remain in district/town homes — exactly the scenario our solution serves.

3 Problem Statement & Target Beneficiaries

Problem Statement (draft): Families of elderly and chronically ill patients lack an affordable way to keep their loved ones safe and medically monitored between hospital visits. This leads to missed medicines, late recognition of danger, and avoidable emergencies. Existing monitoring solutions are costly, English-only, data-siloed, and do not automatically connect families with emergency services — leaving most Indian families dependent on expensive hospital care.

Beneficiary	How SanjivanAI helps
Elderly (60+) living alone or with working family	24×7 monitoring, medicine reminders, and automatic escalation if readings turn dangerous.
Chronically ill patients (BP, diabetes, heart, post-stroke)	Daily vitals tracking, trends visible to the doctor/family, early-warning "danger score".
Family caregivers in cities & villages	See vitals and alerts from anywhere; confirm & act with one tap; no need to be physically present 24×7.
Community health workers / PHC staff	A simple dashboard and alert trail for outreach, home visits and follow-up.
State health authorities	Anonymised, privacy-safe trends for preventive-care planning (future phase).

4 Our Solution: SanjivanAI

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SanjivanAI is a **personal AI nurse that lives in every home** — an affordable, family-first health-monitoring system with three promises:

Promise	What it does
Monitor 24×7	An easy At-Home Monitor / wearable continuously tracks heart rate, blood oxygen (SpO ²), blood pressure, temperature and (later) falls — no nurse or hospital bed required on-site.
Medicine on time	A smart medication manager reminds at the right time, confirms “taken/skipped”, and notifies the family if a dose is missed — stopping a silent killer: missed medicines.
AI that warns early	A transparent rule-based “AI nurse” watches all readings, computes a clear Safe / Caution / Danger score, and explains in simple words why it is worried.
Family in the loop	A clean family dashboard on phone & computer shows live numbers, trends, alerts and medication status — in Bengali, Hindi and English, and readable even by non-technical users.
Auto emergency help	If a reading crosses danger limits: → family alerted first → if not confirmed/acknowledged → the system automatically escalates to ambulance/hospital/108 & helplines (e.g. tele-MANAS) — turning minutes into seconds.

How it works (demo flow)

- 1. Register the patient, add family caregivers and medicines with times.
- 2. The At-Home Monitor (prototype: simulation engine + manual entry; real device in roadmap) streams vitals.
- 3. The AI nurse scores every reading in real time.
- 4. Medicine reminders fire on schedule; results go to the family.
- 5. Any “Danger” event starts the escalation chain: family → emergency contacts → services; the event forms a medical trail for the doctor.

5 Prototype Scope: What is Real vs Simulated (Honest Plan)

Judges value an honest, working demo over an over-claimed one. Our submission will clearly mark the boundary.

Feature	In working prototype	Notes
Dashboard (live vitals, trends, alerts)	Fully real	Responsive web app — works on phone + computer browsers.
Medication manager & on-time reminders	Fully real	Schedule creation, reminders, taken/skipped tracking, missed-dose alert to family.
AI danger-score rules engine	Fully real	Transparent thresholds (BP, pulse, SpO ² , temp) — explainable, no black box.
Family dashboard & roles	Fully real	Multiple caregivers; per-role views; mobile-first.
Emergency escalation chain	Fully real (flow)	Alert family → Acknowledge → auto-call to configured emergency contacts; SMS/email/WhatsApp demonstration.
Signal “108 / ambulance” integration	Simulated	Simulated call ticket + escalation card; real integration is a pilot-phase item.
Vital sensing from hardware	Simulated	Built-in simulator generates realistic readings + manual entry; a low-cost ESP32 + pulse-oximeter hardware path is in the roadmap.

Payments / subscriptions	Simulated	Real plan-pages and pricing; actual payment gateway in pilot.
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Languages	Fully real (English + Bengali/Hindi labels)	Vernacular-first UI for elderly/family.

Safety & ethics disclaimer (kept in app and submission): SanjivanAI is **not** a medical device, does not diagnose, and never replaces a doctor. It supports families and caregivers; all alerts offer a “human-in-the-loop” confirmation, and clinical validation is required before real deployment.

6 Technology Stack (v1 Prototype)

Layer	Choice	Why
Front-end	Responsive web app (HTML/CSS/JS) as a Progressive Web App	One build works on all phones/PCs, no app-store friction; judges click it live; later wrappable into an Android APK.
Back-end	Node.js + Express	Lightweight, easy to host and read for a student team.
Data	SQLite / JSON store	No server cost; portable; easy demo data.
Alerting	Email + SMS + WhatsApp APIs (demonstration endpoints)	Proves real escalation; free/low-cost tiers for demo.
Simulator	Built-in vitals simulator + manual entry	Realistic demo without hardware during evaluation.
Repo & license	Public GitHub/GitLab · MIT	Meets HSC open-source + README/architecture/documentation requirement.

7 Sustainability & Business Model

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- **Affordable subscription — “an AI nurse for every family.”** Proposed tiers: Family plan ~₹99/month, Caregiver+ plan ~₹199/month (multi-member, report sharing), Institutional (PHC/old-age homes) pricing on request — a fraction of today’s ₹20,000+ RPM packages.
- **Public-health convergence:** designed to plug into Ayushman Bharat/ABHA digital health IDs, PHC follow-up workflows and state health missions (future phase), enabling possible empanelment rather than pure B2C.
- **Roadmap to hardware:** a “Made in India” low-cost monitor (ESP32 + pulse-oximeter + BP cuff) targeted under ₹5,000 for rural homes; the app is device-agnostic first.

8 Extra Winning Points (our edge before we even build)

1. Two themes at once — intentional

Our entry squarely addresses **Healthcare, Wellbeing & Service Delivery** and **Elderly Care & Healthy Ageing**. Deliberately mapping a solution to the theme chart shows judges maturity.

2. Digital-Public-Infrastructure alignment

Designed for ABHA integration, tele-MANAS (14416) mental-health linkage, and convergence with Ayushman Bharat PHC follow-up — the vision of "Viksit Bharat."

3. Digital Inclusion — not just a health app

Bengali/Hindi/English UI, a deliberately simple elderly-friendly design, offline-first behaviour, and **SMS/feature-phone fallback** for family alerts where internet is weak — this addresses the "last mile" the guidelines reward.

4. Privacy & trust as a feature (DPDP-aligned)

Explicit consent, patient-control of data, no ads, data minimisation, India-first storage plan — designed from day one to meet the Digital Personal Data Protection expectations judges increasingly probe.

5. Climate & community resilience angle

Heatwave/humidity-aware alerts (e.g. elderly during Bengal summers) tie the health solution to disaster-resilience too — a differentiator few teams will show.

6. Women-led & inclusive caregiving design

Designed around the reality that most unpaid caregiving is done by women — and the guidelines prefer a female team member & a non-tech member. Our content/video lead can be from a healthcare/social-science background.

7. Honesty that scores with judges

Clear "real vs simulated" boundaries, medical disclaimers, human-in-the-loop confirmation before any emergency action, and a full AI-usage disclosure (as rules require) — credibility is a differentiator.

8. Measurable impact story

Demo metrics: e.g. "adherence up from <50% to 90%", "response time to danger reading from ~hours to seconds", "cost of monitoring cut by >80% vs hospital."

9. SDG alignment

Explicitly mapped to UN SDG 3 (Good Health & Well-Being), SDG 9/10 (innovation & reduced inequalities) — shown on one slide.

10. Built to be open source

MIT-licensed, clean architecture, README + sample data — matching the repository requirements and encouraging state-level reuse.

9 Mapping to the Evaluation Matrix

Evaluation parameter	How SanjivanAI answers it
Relevance	Direct match to two listed thematic areas; state-specific (West Bengal ageing + rural) problem statement.
Technical Strength	Clean responsive PWA + Node backend + documented rules engine + simulator; open-source MIT repo with README & architecture.

Functionality	Live clickable dashboard: vitals, medicines, alerts, escalation — demonstrated end-to-end in the video.
Creativity / Innovation	"AI nurse at home" model: affordability + vernacular + SMS fallback + dual-theme positioning.
Social Cause Impact	Concrete metrics story (adherence, response time, cost); elderly + rural beneficiaries; SDG 3.
Presentation & Team	Role-based team (lead / dev / healthcare-content), 3–5 min narrative video, 6–7 slide deck with AI disclosure.

10 Submission Checklist & Naming Convention

Files must be named **<State>_<TeamName>_<Document>.pdf** (e.g. WestBengal_TeamX_ProblemStatement.pdf).

Deliverable	Specification	Status
Problem Statement	State-specific problem, beneficiaries, current gaps.	In progress
Presentation Deck (6–7 slides)	PPT/PDF, ≤10 MB; problem → solution → impact → team + AI disclosure.	To build
Working Prototype	Public GitHub link: code, README, architecture, sample data, license.	To build
Demo Video (3–5 min)	MP4/MOV, ≥720p, ≤80 MB; team + problem + prototype + impact.	To build
Annexure 1 + IDs	Team details, signed self-cert, self-attested student IDs.	Pending

11 Working Timeline (from today, 08 Sep 2026)

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Dates	Milestone
08–20 Sep	Finalise problem statement & MVP scope; build prototype v0 (dashboard + simulator + medicines).
20 Sep – 05 Oct	Add AI danger engine + family roles + escalation flow; local-language labels; polish UI for demo.
05–10 Oct	Write README/architecture, sample data, push open-source repo; prepare 6–7 slide deck.
10–14 Oct	Record 3–5 min demo video (≥720p); collect Annexure-1 + ID proofs; final review.
On/before 15 Oct	Team Lead submits everything on MY Bharat Portal.
23–05 Nov	State-level hackathon & demonstration (mentoring + jury).
01–15 Dec	IIT Bombay screening to national finalists (36 teams).
10–12 Jan 2027	National Showcase, New Delhi — stall, pitch, awards.

12 Team & Roles (to finalise)

Role	Responsibility	Suggested member
Team Lead	Registration, submissions, deck, coordination.	[Name]
Developer	Prototype build, repo, code quality, demo pipeline.	[Name]
Healthcare / Content & Video	Problem statement, medical accuracy, video narration, local outreach.	[Name — ideal: female / social-science]

Open action items (next session)

- Decide final team name (appropriate, non-derogatory; e.g. “Sahaay Tech”).
- Confirm every member is individually registered on MY Bharat portal.
- Confirm AISHE college/institution details before 15 Oct.
- Approve app working title (working: **SanjivanAI**).

AI-Tools Disclosure (required by rules): This concept document and the prototype are being developed with AI-assisted coding (opencode/CLI assistant) under team supervision. The final submission & presentation deck will clearly disclose the extent of AI usage, as mandated for HSC 2027.